Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
  - Customer demographics (Age, Gender, Location, Subscription Status)
  - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
  - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.
- **Missing Data Handling:** Checked for null values and imputed missing values in the `Review Rating` column using the median rating of each product category.
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
  - Created **age_group** column by binning customer ages.
  - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned

DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in Microsoft SQL Server Management Studio to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender	revenue
1	Male	157505
2	Female	73995

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id	purchase_amount
1	7	74
2	8	68
3	18	87
4	36	72
5	45	61
6	55	75
7	56	80
8	58	89
9	62	81
10	63	80
11	64	76

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_purchased	Average Product Rating
1	Trench Coat	3.99
2	Loafers	3.91
3	Bracelet	3.89
4	Sneakers	3.86
5	Backpack	3.85

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_type	(No column name)
1	Standard	59
2	Express	59

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status	total_customers	avg_spend	total_revenue
1	Yes	1020	58	59867
2	No	2880	59	171633

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased	discount_rate
1	Windbreaker	57.140000000000
2	Heels	53.330000000000
3	Fleece	52.500000000000
4	Gloves	51.720000000000
5	Hat	51.280000000000

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment	Number of Customers
1	Returning	726
2	Loyal	3106
3	New	68

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

	item_rank	category	item_purchased	total_orders
1	1	Accessories	Watch	93
2	2	Accessories	Earrings	92
3	3	Accessories	Sunglasses	89
4	1	Clothing	T-shirt	131
5	2	Clothing	Shorts	130
6	3	Clothing	Jacket	118
7	1	Footwear	Oxfords	73
8	2	Footwear	Sneakers	70
9	3	Footwear	Slippers	66
10	1	Outerwear	Trench Coat	46
11	2	Outerwear	Poncho	43
12	3	Outerwear	Parka	42

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

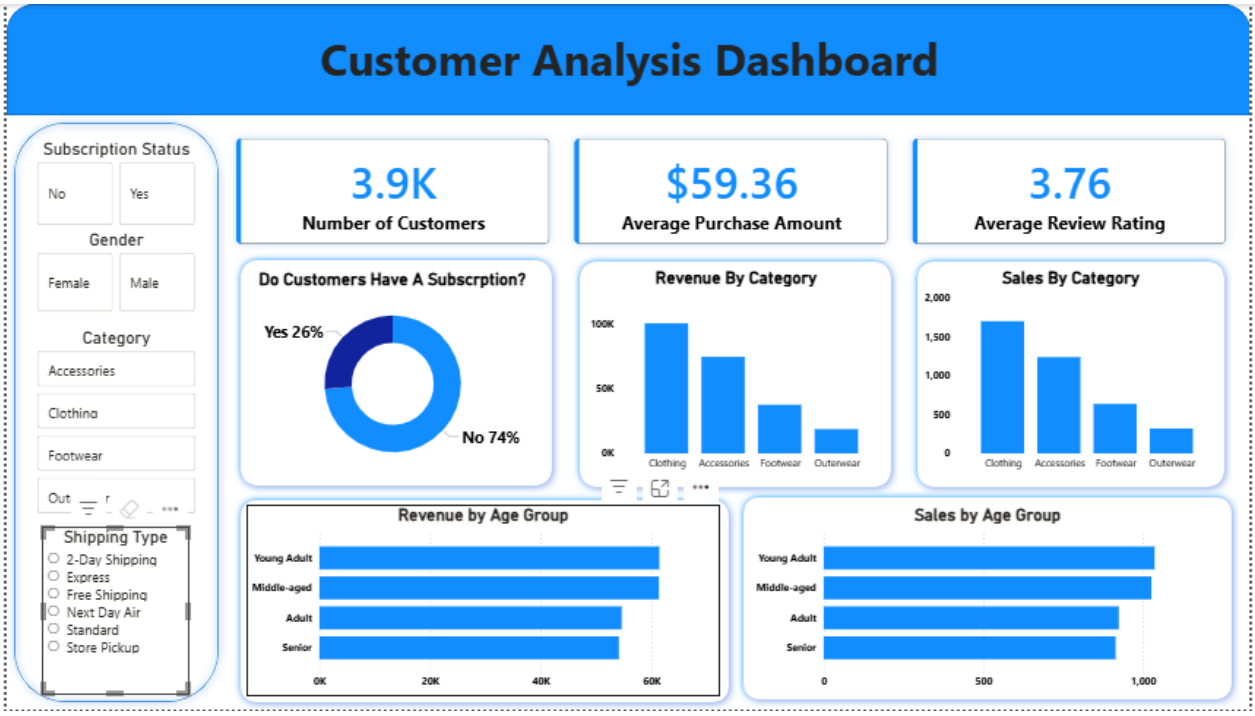
	subscription_status	repeat_buyers
1	Yes	916
2	No	2598

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

	age_group	total_revenue
1	Young Adult	61416
2	Middle-aged	61351
3	Adult	54622
4	Senior	54111

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.